# QKD Eavesdropping Detection with Machine Learning

## Results and plots explained — BB84, BKM07 and E91

Source: branch *claude/wonderful-meitner-16r3p0* of himeshsoni704/QuantumKeyDistribution (latest commit at time of writing, 4 Oct 2026), notebook *QuantumKeyDistribution_Draft2.ipynb*, its *plots/* and *data/* folders.

**Read this first (honest caveats).** (1) The notebook was re-run after the Draft 2.1 bug-fix commit, but its saved text outputs stop at cell 158 (deep learning headline experiment); later results (CUSUM, minimum detectable strength, calibrated benchmark, SHAP, bursty/mixed attacks, grouped DL CV) are taken from the CSV files and plots committed in *data/* and *plots/*. The branch was still receiving automatic checkpoint commits, so numbers may change on the next run. (2) The main BB84/BKM07/E91 datasets were loaded from cached CSVs generated before the fix pass, so they do not use the unified K margin. (3) Many results sit at AUC = 1.0, which means the task is too easy to rank methods — that is itself a finding and is flagged wherever it occurs.

# 1. What the project asks

In Quantum Key Distribution an eavesdropper (Eve) must disturb the channel, but real channels are noisy, so a raised error rate is ambiguous. The project simulates three protocols — BB84 (prepare-and-measure, fully quantum), BKM07 (semi-quantum: Bob can only measure in Z or reflect) and E91 (entanglement-based) — with realistic noise and several attacks, turns every simulated run into a feature vector, and asks how well classical ML and deep learning separate ‘secure’ from ‘Eve present’. AUC (area under the ROC curve; 0.5 = coin-flip, 1.0 = perfect ranking) is the main metric; FNR is the fraction of attacks missed.

# 2. Physics sanity checks

**Channel model.** The closed-form fibre model (GYS calibration) gives QBER 3.30% at 0 km rising to 3.76% at 100 km and 4.6% at 122 km (the real experiment measured 8.9% there, so the model is only trusted to about 100 km). Gain falls from 2.1e-2 at 0 km to 1.7e-4 at 100 km. The vectorised simulator reproduces theory: at 25 km, gain 6.38e-3 vs 6.43e-3 theory, QBER 0.0347 vs 0.0331.

**E91 Werner state.** Measured CHSH |S| tracks $2\sqrt{2}\cdot V$ (2.706 vs 2.687 at V = 0.95) and key QBER tracks (1–V)/2 (0.0256 vs 0.025). Intercept-resend on BB84 at 100% intensity gives QBER 0.435 (about 0.5 minus noise), 50% gives 0.26 — as expected.

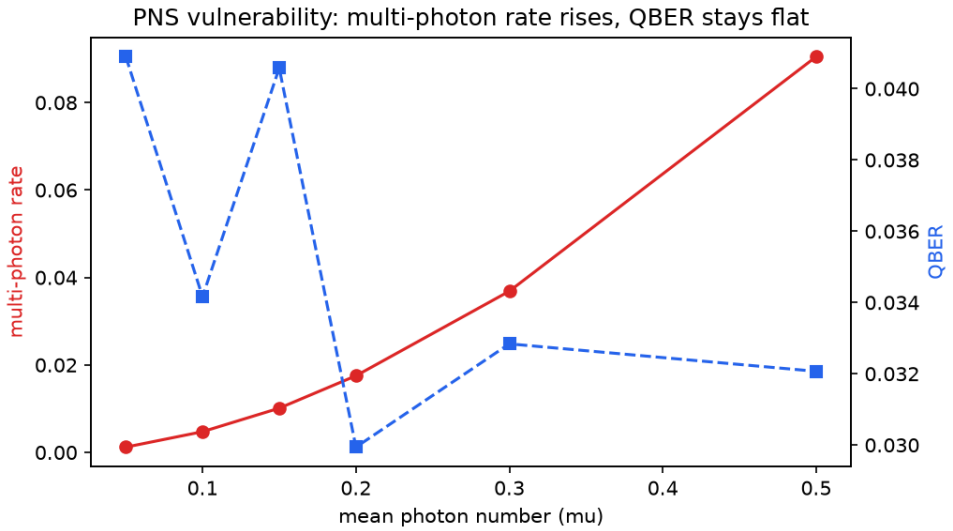



Fig. PNS vulnerability. Red: multi-photon rate grows with mean photon number μ (0.1% at μ=0.05 to 9% at μ=0.5). Blue: QBER measured with the simulator under a full-time PNS attack stays around 0.030–0.041 (noise from ~500 sifted bits), i.e. PNS adds no errors — which is why QBER alone cannot see it.

# 3. How many pulses are enough? (convergence plots)

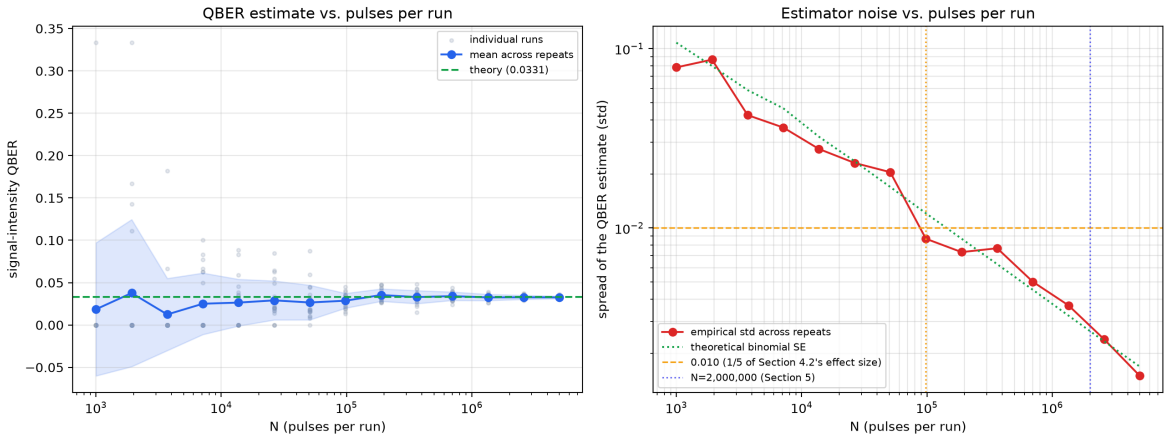



Fig. QBER estimate vs pulses per run (BB84, 25 km). Left: individual runs scatter wildly below ~10^4 pulses and converge to the theory line (0.0331). Right: the empirical spread follows the binomial standard error; it drops below the 0.01 target (one fifth of the 5-point attack effect) at ~10^5 pulses.

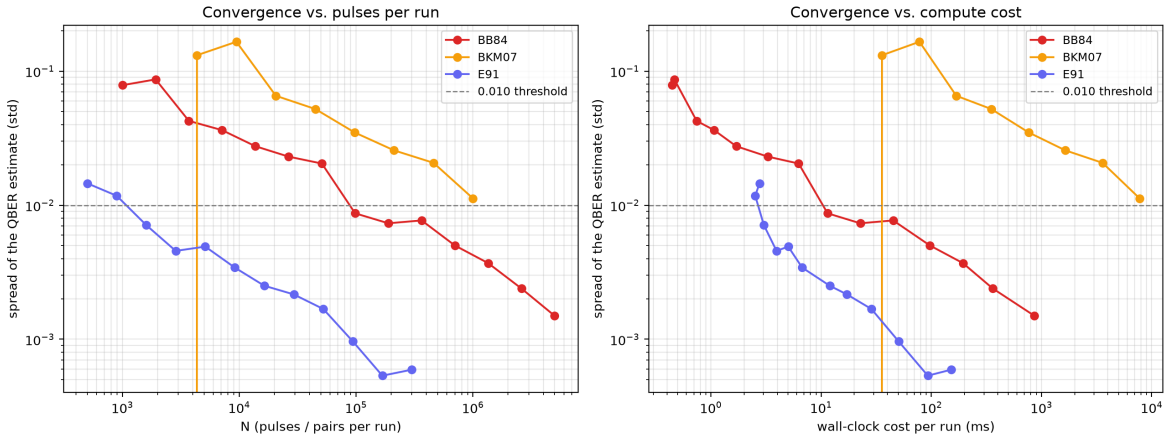



Fig. Cross-protocol convergence. E91 needs only ~1,600 pairs (no loss), BB84 ~98,000 pulses, and BKM07 did not reach the target even at 1,000,000 pulses because round-trip loss leaves ~0.05% of pulses as usable key rounds. Right panel converts this to wall-clock: BKM07 is by far the most expensive.

# 4. Leakage audit (are the labels secretly given away?)

Eight tests (L-1 single-feature AUC, L-2/3 constants and disjointness, L-4 nuisance KS test, L-5 equal sample length, L-6 NaN rates, L-7 shuffled labels, L-8 zero-strength control) were run on all three protocols.

| Protocol | Findings |
|---|---|
| BB84 | No leak flags. Strongest single feature r_secure AUC 0.945 (<0.99). L-5: N sets identical across classes (up to the 20M-pulse cap). L-7 shuffled AUC 0.496, L-8 zero-strength AUC 0.528 — both ~0.5, so the label does not leak. |
| E91 | No flags. Real per-run N = 11,700 (the earlier audit mistakenly used 5,000). L-7 0.506; L-8 0.415 (within tolerance). |
| BKM07 | L-1 flags 7 features above 0.99 (qber_zc 1.000, h_ctrl 1.000, ctrl_sift_ratio 1.000) and qber_zc classes are disjoint. L-7 0.549 and L-8 0.483 pass. Interpretation: attacks are strong relative to the ~0.004 standard error of CTRL-round QBER at K=2,000, so this is genuine signal, not label leakage — but it means the BKM07 task is trivially easy. |

# 5. Classical ML results (Sections 8–11)

Models: KNN, logistic regression, tuned random forest, tuned SVM-RBF, tuned XGBoost, and an unsupervised Isolation Forest trained only on secure runs. Hyper-parameters were tuned with group-aware 5-fold CV so that channel-matched ‘twin’ runs never straddle folds. Test set = 20% of runs, split by run index.

| Protocol | KNN | LogReg | RF | SVM | XGBoost | IsoForest |
|---|---|---|---|---|---|---|
| BB84 AUC | 0.9967 | 0.9986 | 0.9999 | 0.9989 | 0.9997 | 0.8747 |
| BKM07 AUC | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9993 |
| E91 AUC | 0.8698 | 0.9069 | 0.9162 | – | 0.9206 | 0.8696 |

BB84 is almost perfectly separable and BKM07 is perfectly separable by every model including the unsupervised one. E91 is the hard case (AUC 0.87–0.92; FNR about 10–14%, FPR about 30–40% at threshold 0.5). Because ceiling results cannot rank models, the more informative experiments follow.

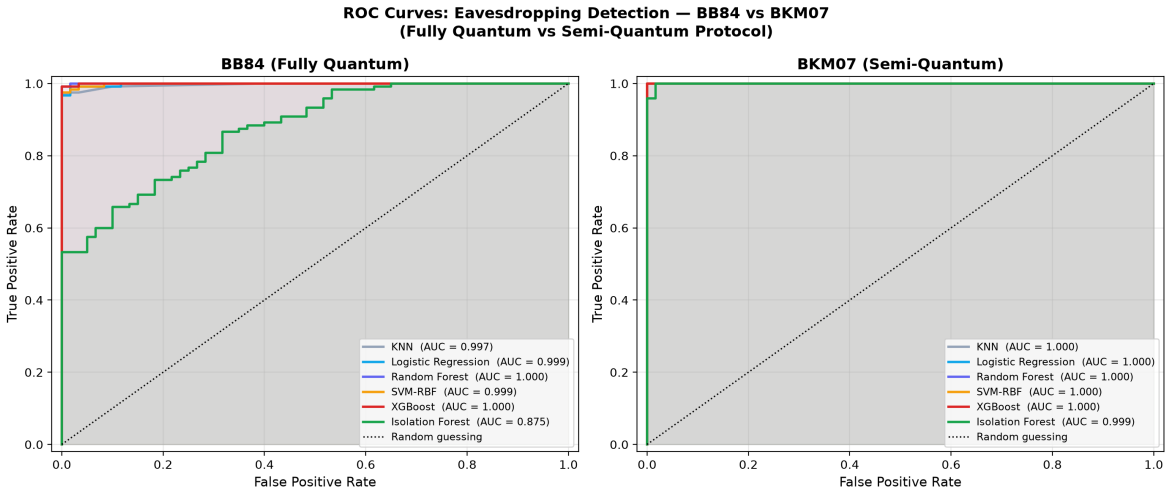



Fig. ROC curves for BB84 (left) and BKM07 (right). Supervised models hug the top-left corner; the Isolation Forest (green) is clearly weaker on BB84 (AUC 0.875) but nearly as good on BKM07.

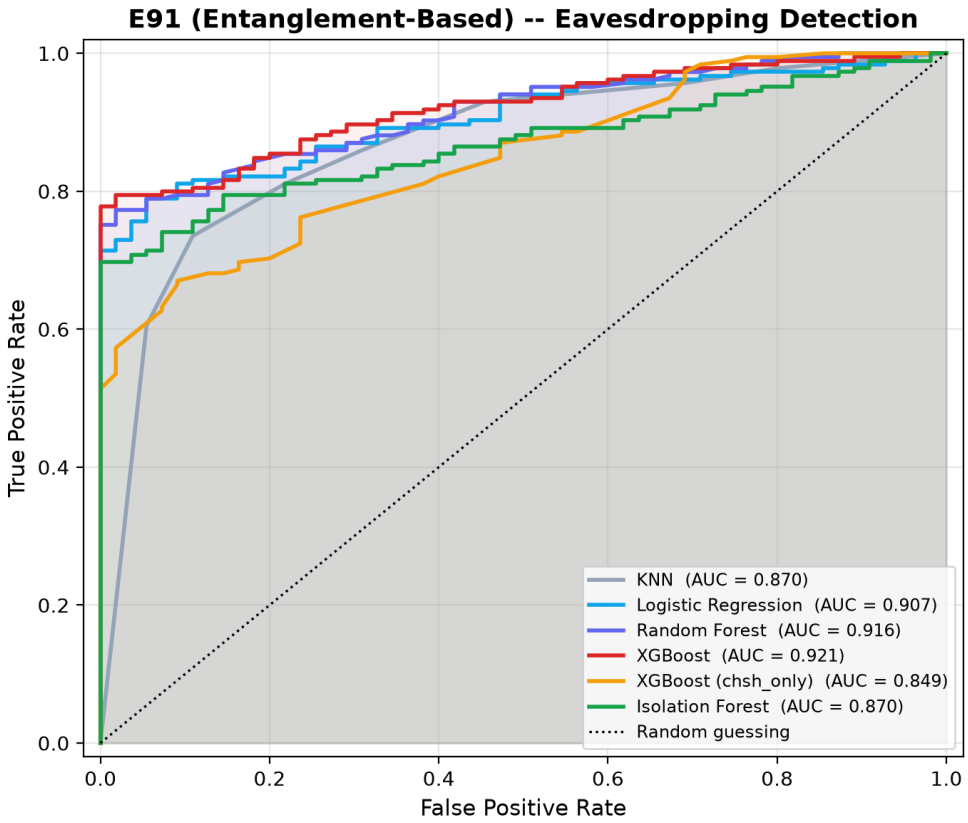



Fig. E91 ROC. XGBoost (0.921) and random forest (0.916) lead; the single-feature CHSH-only model (orange, 0.849) is the 'standard Bell test' baseline. The 11 physics features add +0.07 AUC; the gain is concentrated on the ancilla attack (FNR 0.151 → 0.038) while intercept-resend and loss manipulation are not better detected by the extra features (FNR 0.081 vs 0.065; 0.157 vs 0.129).

# 6. Which information drives detection? (ablation, importance, SHAP)

**Ablation (20 splits, 95% CI):** A = QBER only, A+ = plus aggregate statistics, B = plus temporal features, C = plus decoy-state features.

| Group | AUC [95% CI] | FNR @1% FPR | FNR pns |
|---|---|---|---|
| A QBER only | 0.746 [0.731, 0.761] | 0.527 | 0.982 |
| A+ aggregate | 0.743 [0.732, 0.754] | 0.523 | 0.984 |
| B + temporal | 0.734 [0.721, 0.747] | 0.530 | 0.987 |
| C + decoy (all) | 0.997 [0.995, 0.998] | 0.026 | 0.013 |

**Key result:** the whole jump (0.74 → 0.997) comes from the decoy-state features, because PNS adds no QBER and is invisible without them (98% of PNS attacks missed by groups A–B). **Temporal features (B) add nothing** — their AUC is slightly lower than A+ with overlapping intervals — so the notebook does not establish that temporal structure helps. Re-seeding under three master seeds gives the same ranking (C best, range of C only 0.0005).

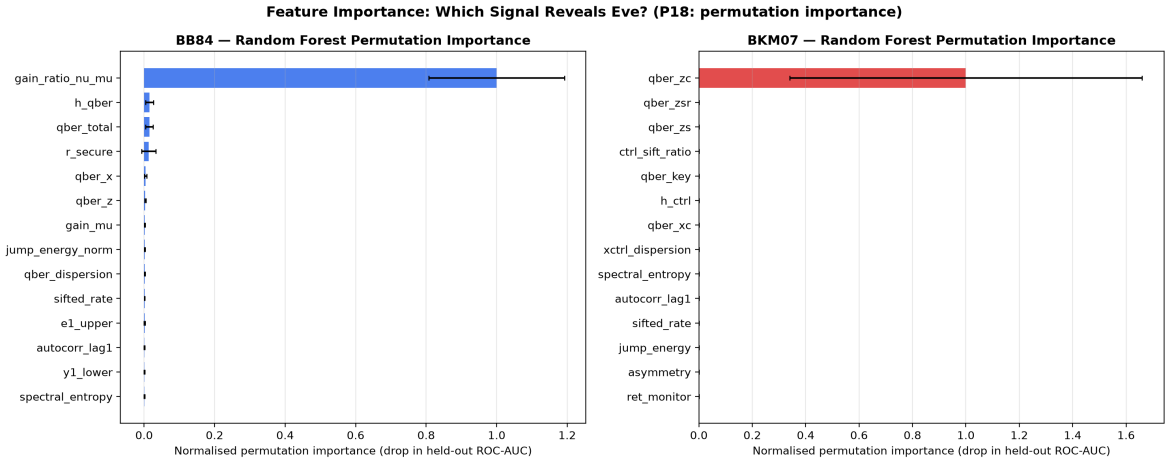



Fig. Permutation importance (random forest). BB84 is dominated by gain_ratio_nu_mu (the decoy/signal gain ratio that PNS distorts); BKM07 by qber_zc (the CTRL-round error rate).

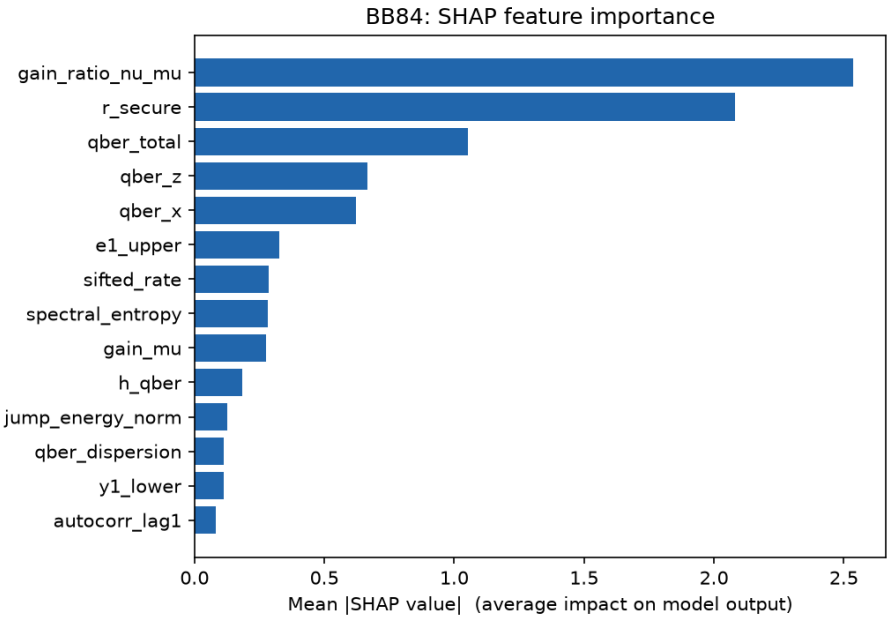



Fig. SHAP, BB84: gain_ratio_nu_mu (2.5) and r_secure (2.1) dominate, then qber_total (1.05). Temporal features are small, consistent with the ablation.

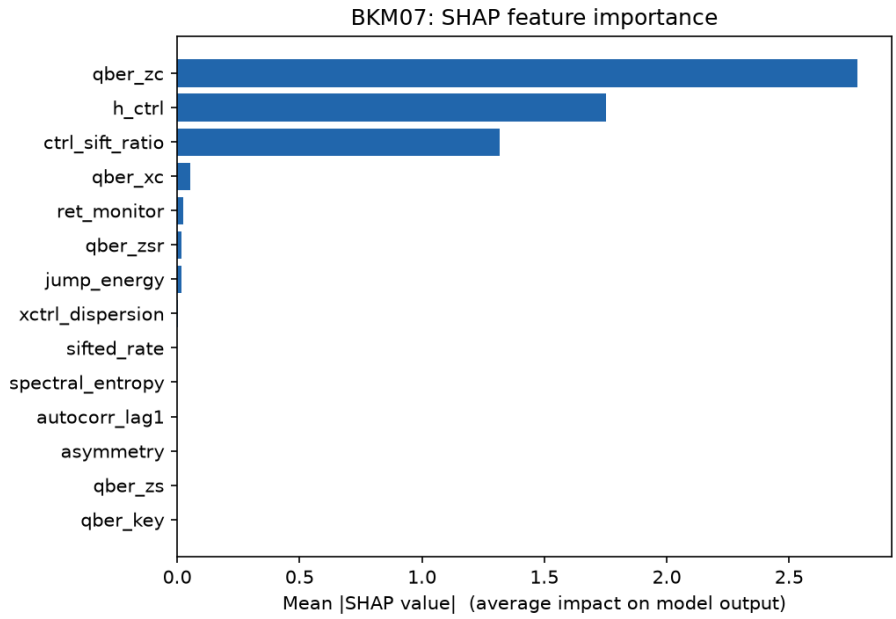



Fig. SHAP, BKM07: only three CTRL features matter (qber_zc 2.8, h_ctrl 1.75, ctrl_sift_ratio 1.3); everything else is ~0. The reflected-photon check is the entire signal.

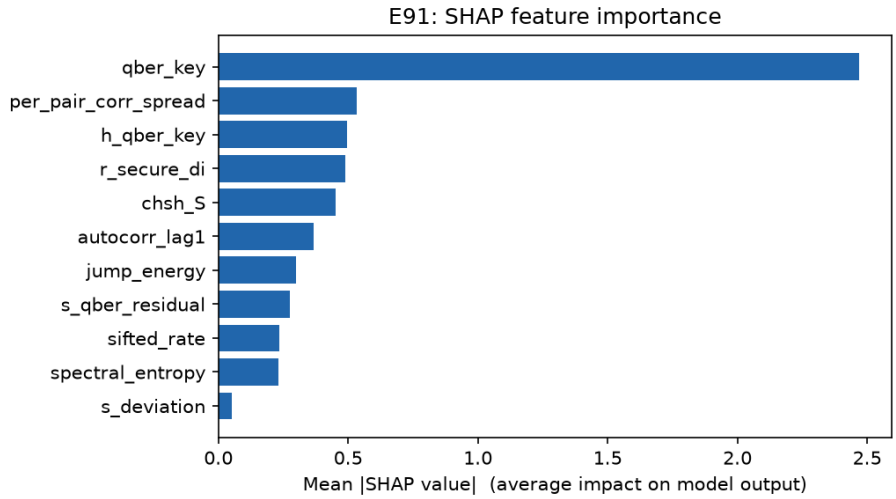



Fig. SHAP, E91: qber_key (2.5) dominates; per_pair_corr_spread, h_qber_key, r_secure_di and chsh_S follow. s_qber_residual contributes only 0.28 — it mainly helps on the ancilla attack.

# 7. Does it generalise? (E2–E4)

These split by attack strength or distance so the test region is unseen in training.

| Test | Result | Meaning |
|---|---|---|
| E2 weak → strong intercept-resend | AUC 0.985, FNR 0.108 | Training on weak attacks still catches strong ones. |
| E2 strong → weak (hard) | AUC 0.682, FNR 0.769 | Weak attacks (≤0.10 intensity) hidden by noise are mostly missed when only strong ones were seen. Only 17 weak runs exist, so this is noisy. |
| E3 0–40 km → 60–100 km | AUC 0.986 | Handles unseen distance (a pure QBER threshold would not). |
| E3 60–100 → 0–40 km | AUC 0.989 | |
| E4 leave-one-cell-out | Cells scored: AUC 0.96, 0.97, 0.83, … | Only cells containing clean runs (intensity bin 0) can be scored; the rest are single-class. |

# 8. Sample size and window sensitivity

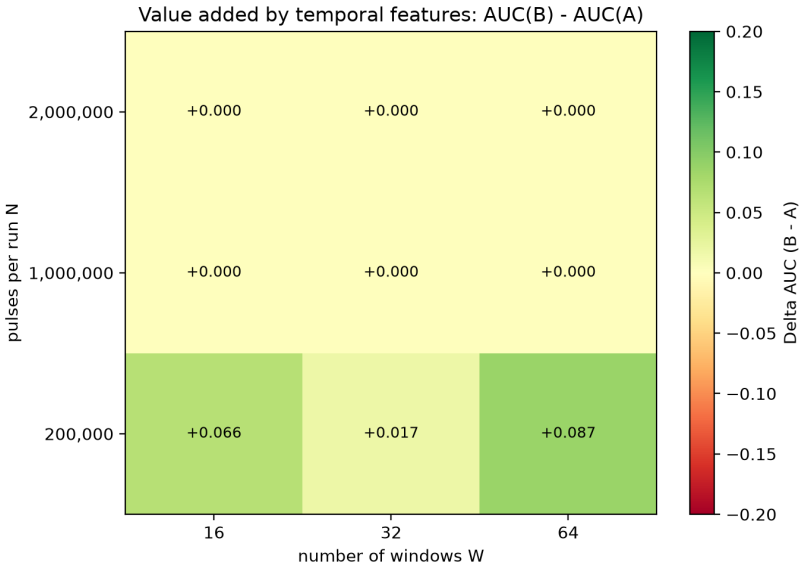



Fig. Gain from temporal features, ΔAUC = AUC(B) – AUC(A), for run length N and number of windows W. At 1M and 2M pulses both models are already at AUC 1.0, so the gain is 0. Only at N = 200,000 does a gain appear (+0.017 to +0.087) — and with 40 runs per class on a single split this is noisy.

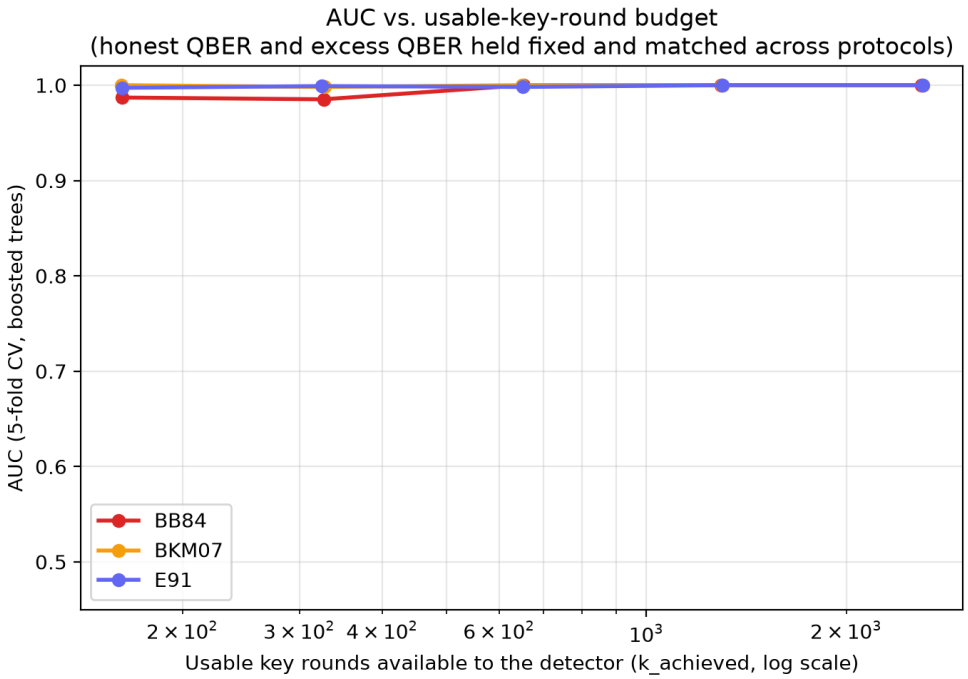



Fig. AUC vs usable key rounds. At matched noise and a +0.05 excess-QBER attack all three protocols reach AUC ≥ 0.985 already with ~160 key rounds. BB84 is slightly lowest at the smallest budget (0.987 vs 1.000 and 0.997).

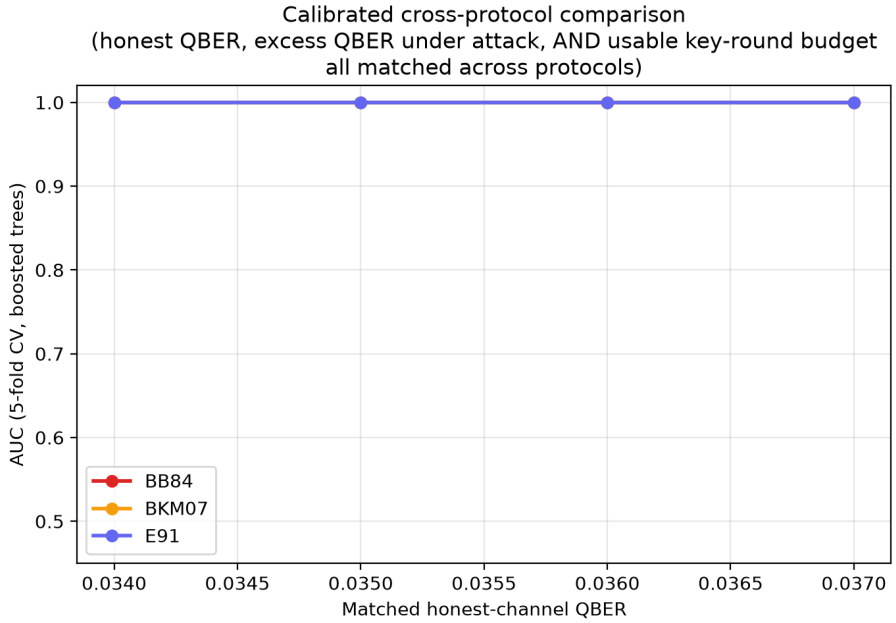



Fig. Calibrated cross-protocol benchmark: honest QBER 0.034–0.037, same excess QBER, matched key budget (k = 2,000 down to 1,068). Every point is AUC = 1.0, so this benchmark is saturated and cannot rank the protocols. Note also that different attacks were used per protocol (BB84 intercept-resend, BKM07 symmetric, E91 ancilla). Lowering the excess-QBER target is needed for a meaningful comparison.

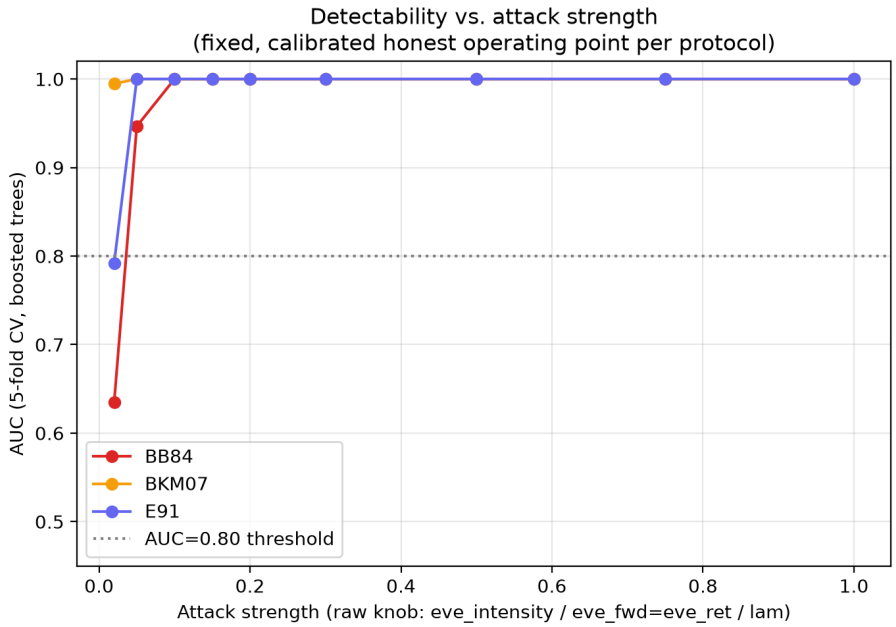



Fig. Minimum detectable attack strength. At strength 0.02 AUC is BB84 0.63, E91 0.79, BKM07 0.99; at 0.05 BB84 0.95, E91 and BKM07 1.0. All protocols exceed AUC 0.8 by strength 0.05, so at that operating point the weakest detectable attack is very small; BB84 is the hardest, BKM07 the easiest.

# 9. How fast does it react? (CUSUM)

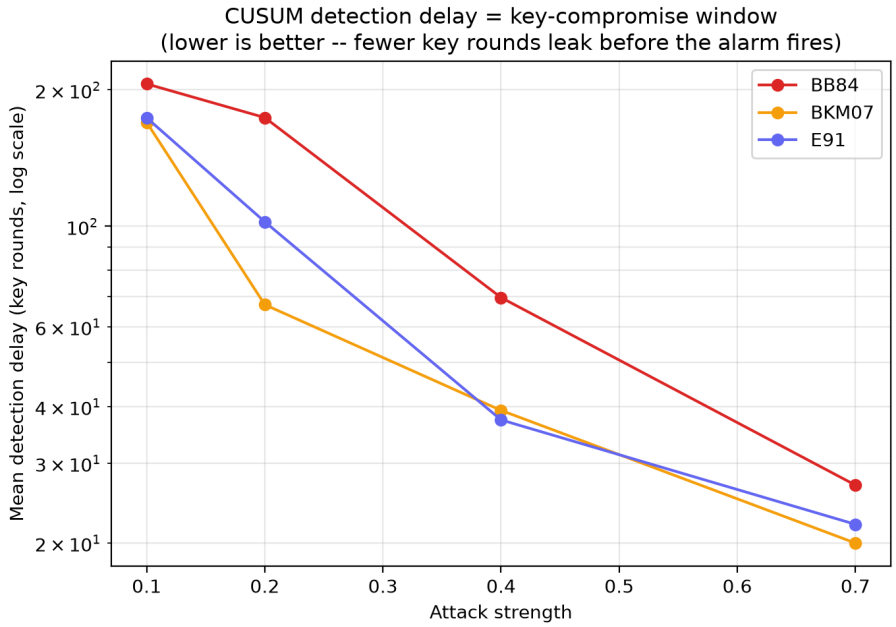



Fig. Sequential CUSUM detection delay in key rounds (lower = fewer key bits leak before the alarm). Delays fall with strength for every protocol: at strength 0.7 about 27 (BB84), 22 (E91) and 20 (BKM07) key rounds; at 0.1, about 206, 174 and 170 with 8/20, 3/20 and 3/20 runs missed. BB84 is slowest at every strength. This figure comes from the fixed experiment in which streams are sized by key-round budget (300 honest + 600 attacked); the earlier version gave BB84 and BKM07 almost no key rounds.

## 10. Robustness, mixed attacks, calibration

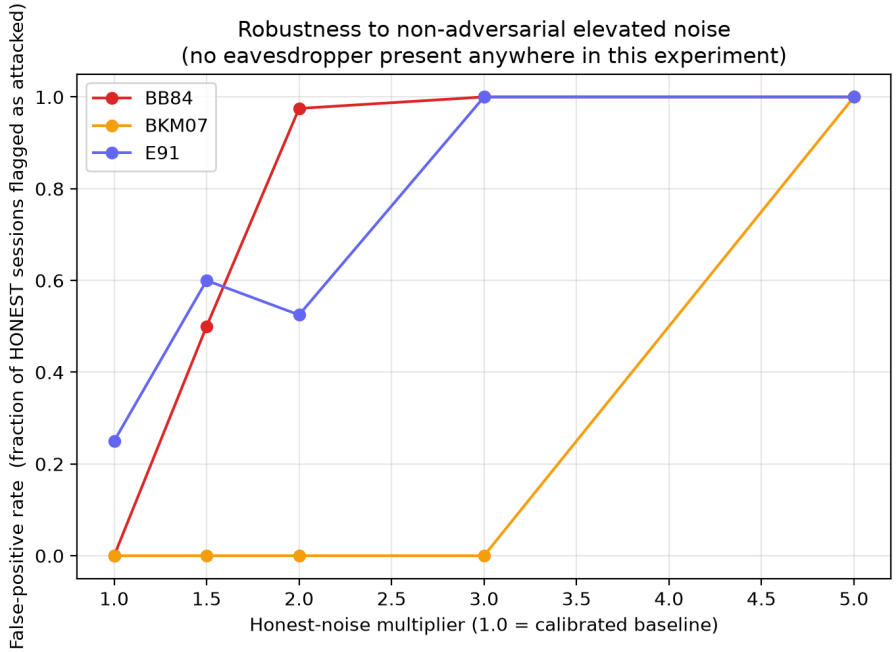



Fig. False-positive rate on completely honest links with elevated noise. A classifier trained on the main dataset starts flagging honest noise: BB84 reaches 50% false positives at 1.5× the calibrated noise and 97.5% at 2×; E91 already 25% at baseline and 60% at 1.5×; BKM07 is clean until 3× (0%) and fails at 5×. A classifier that reacts to ‘noisier than usual’ rather than to an attack signature would need retraining against honest drift before deployment.

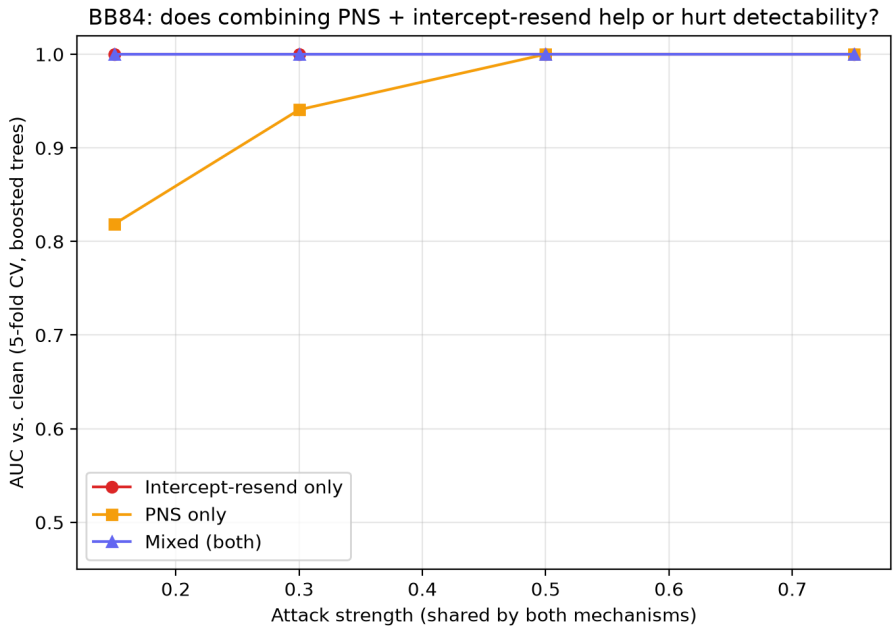



Fig. Mixed PNS + intercept-resend (BB84 only). Mixed AUC is 1.0 at every strength; PNS alone is harder at low strength (0.82 at 0.15, 0.94 at 0.3). Combining attacks does not hide them: they write to different observables (extra errors vs. changed yield), so fingerprints add rather than cancel (mixed minus best single = 0.0).

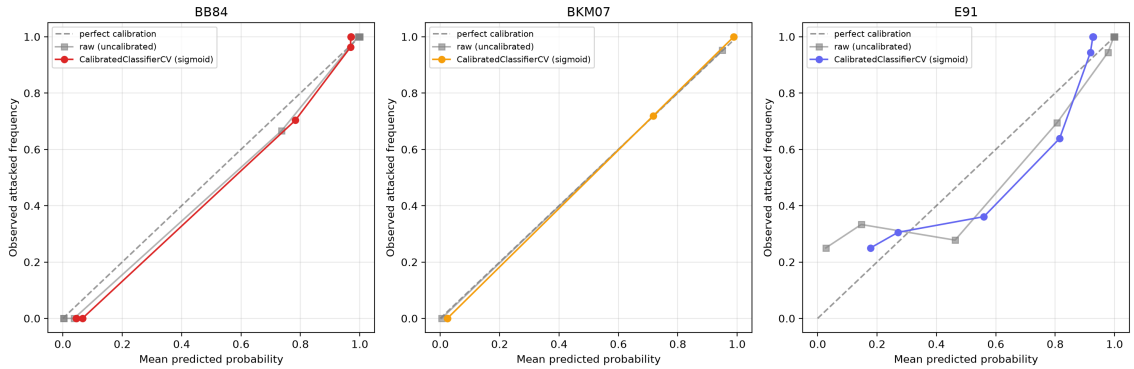



Fig. Reliability diagrams (predicted probability vs observed attack frequency). Platt scaling brings BB84 and BKM07 close to the diagonal; E91 is noisy because its test set is small and its classes overlap.

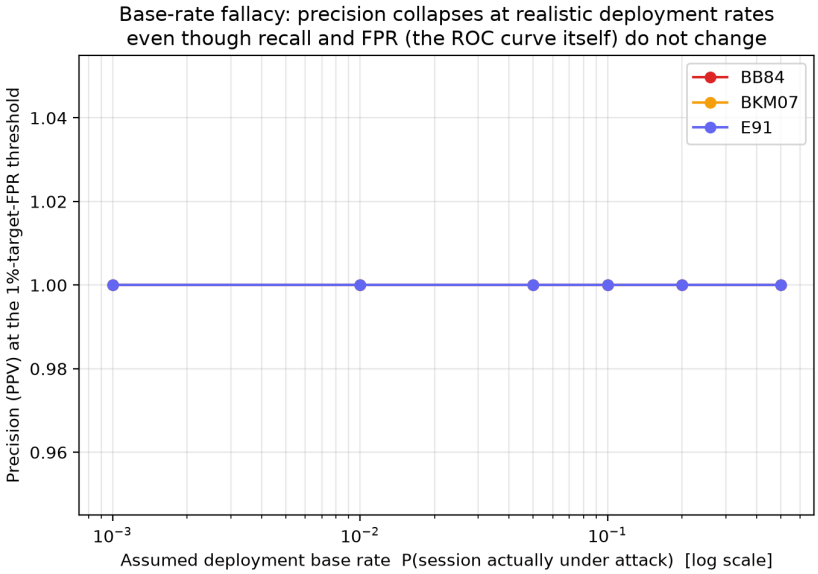



Fig. Precision vs deployment base rate. In the committed run precision is 1.0 at every base rate — the classifier achieved 0 false positives at the 1%-FPR threshold on the test set, so the 'base-rate fallacy' collapse is not visible; with so few honest test sessions the estimate of FPR is itself coarse. Recall at 1% FPR: BB84 0.894, BKM07 1.0, E91 0.744.

# 11. Deep learning (Section 18)

A CNN + BiLSTM + attention network with a per-protocol input adapter and shared trunk sees 96-event windows (8 numbers per event). Experiments: per-protocol attack-type classification, leave-one-protocol-out transfer (pretrain on two protocols with a domain-adversarial head and supervised-contrastive loss, freeze the trunk, train a new adapter on a fraction of the third protocol, compare with training from scratch), Deep SVDD zero-day detection, and 5-fold session-grouped cross-validation.

| Experiment | Result |
|---|---|
| Attack-type classifier (macro-F1) | BB84 0.438, E91 0.534, BKM07 0.903. BB84 confuses clean with blocking / loss manipulation; E91 confuses intercept-resend with loss manipulation. |
| Transfer vs scratch (AUC, full data) | →BB84: 0.661 vs 0.637 (not significant). →E91: 0.916 vs 0.879 (significant, +0.036). →BKM07: 0.911 vs 0.970 (scratch better). |
| Classical-features baseline AUC | BB84 0.985, E91 0.914, BKM07 1.000 — the hand-crafted physics features beat the deep model on BB84 and BKM07 and tie on E91. |
| Grouped 5-fold CV (15 folds) | Overall AUC 0.803 ± 0.030; BB84 0.528 ± 0.108 (about chance), E91 0.812, BKM07 0.966. |

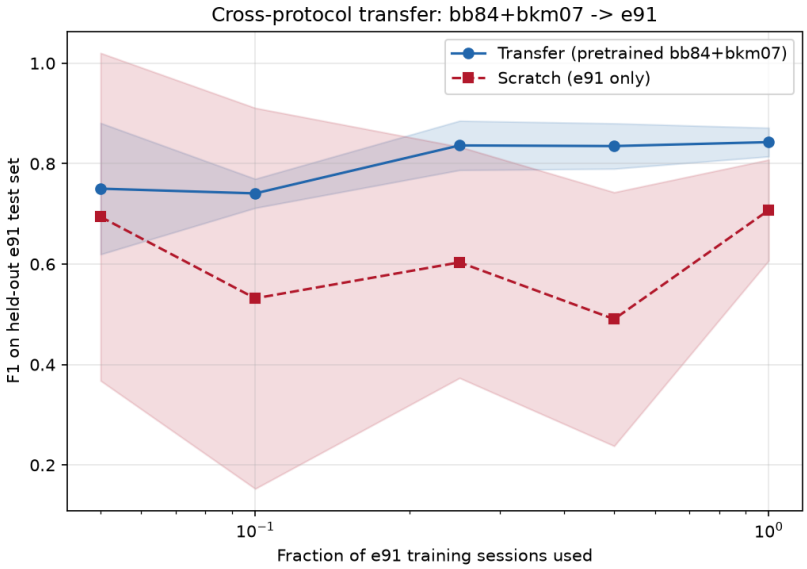



Fig. Transfer into E91. Pretrained (blue) is above scratch (red) at every training fraction in F1 (0.84 vs 0.71 at 100%) — the best evidence for cross-protocol transfer in the notebook.

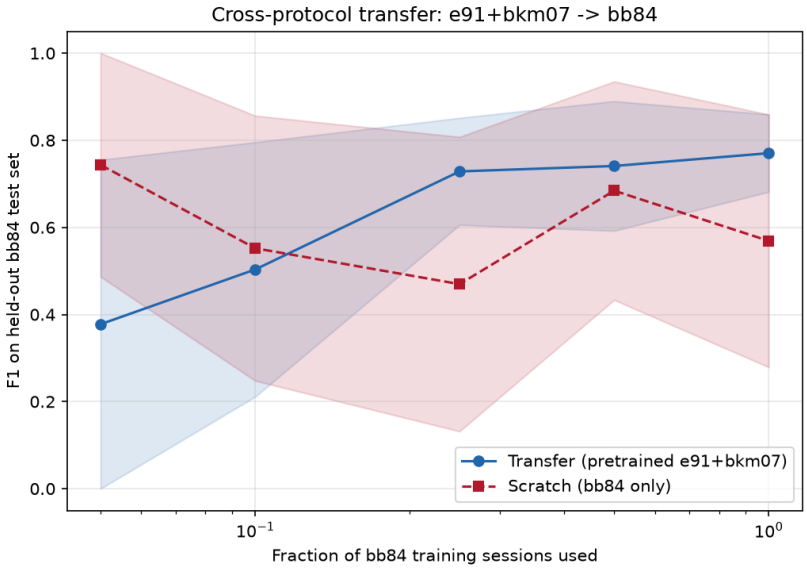



Fig. Transfer into BB84. Transfer F1 rises with more data and beats scratch above 25% of the data, but AUC stays at 0.56–0.66: BB84 windows of ~48 sifted bits contain too little information for the network.

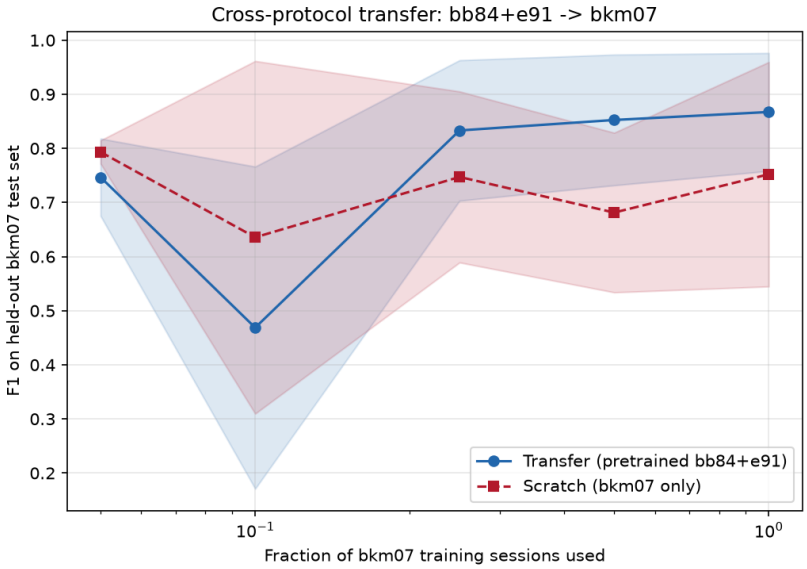



Fig. Transfer into BKM07: wide confidence bands; scratch has the better AUC (0.97 vs 0.91). BKM07 is easy enough that a freshly trained network needs no help.

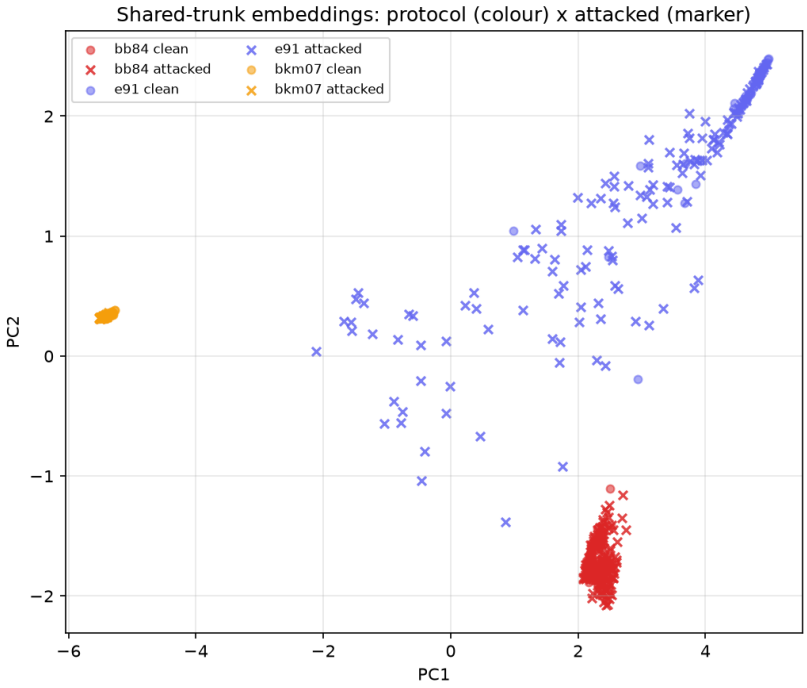



Fig. PCA of the shared-trunk embedding. Clusters are separated by **protocol** (red BB84, blue E91, orange BKM07 clearly apart) and only partly by clean/attacked inside a protocol — E91 shows an attacked arm, BB84 and BKM07 are tight blobs. So the trunk still encodes protocol identity more than attack-ness, limiting transfer.

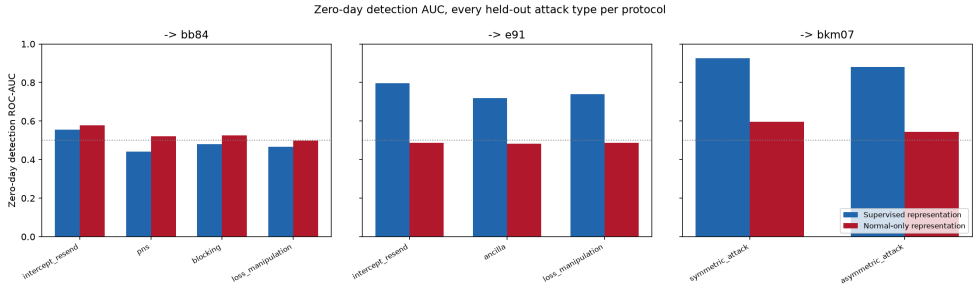



Fig. Zero-day detection with held-out attacks. Supervised-pretrained representation (blue) vs normal-only Deep SVDD (red). BB84: both near 0.5 (chance). E91 and BKM07: supervised 0.7–0.9 vs normal-only ~0.5. The normal-only branch effectively learns nothing, so the gap says supervised beats a non-functioning baseline.

# 12. Bottom line

- **What works:** decoy-state features expose PNS (AUC 0.74 → 0.997); the CTRL error rate exposes every BKM07 attack; QBER plus Bell-test features detect E91 attacks to AUC ~0.92; CUSUM catches strong attacks within 20–30 key rounds in all three protocols.
- **What is weak:** E91 loss manipulation (FNR 0.16); weak intercept-resend when unseen in training (AUC 0.68); classifiers also fire on honest noisy links (false positives rise sharply at 1.5–2× baseline noise).
- **What does not hold up:** temporal features add no measurable value; deep learning is at chance on BB84 and is beaten by the classical features; cross-protocol transfer helps only for E91; Deep SVDD normal-only is at chance.
- **Saturation:** BKM07 main task and the calibrated benchmark / min-strength / mixed-attack experiments sit at AUC 1.0, so they cannot rank protocols; use weaker attacks (smaller target excess QBER) before drawing protocol comparisons.
- **Scope:** individual per-pulse attacks only, asymptotic key-rate features, no detector-side or collective attacks, simulation only; results are not a security proof.